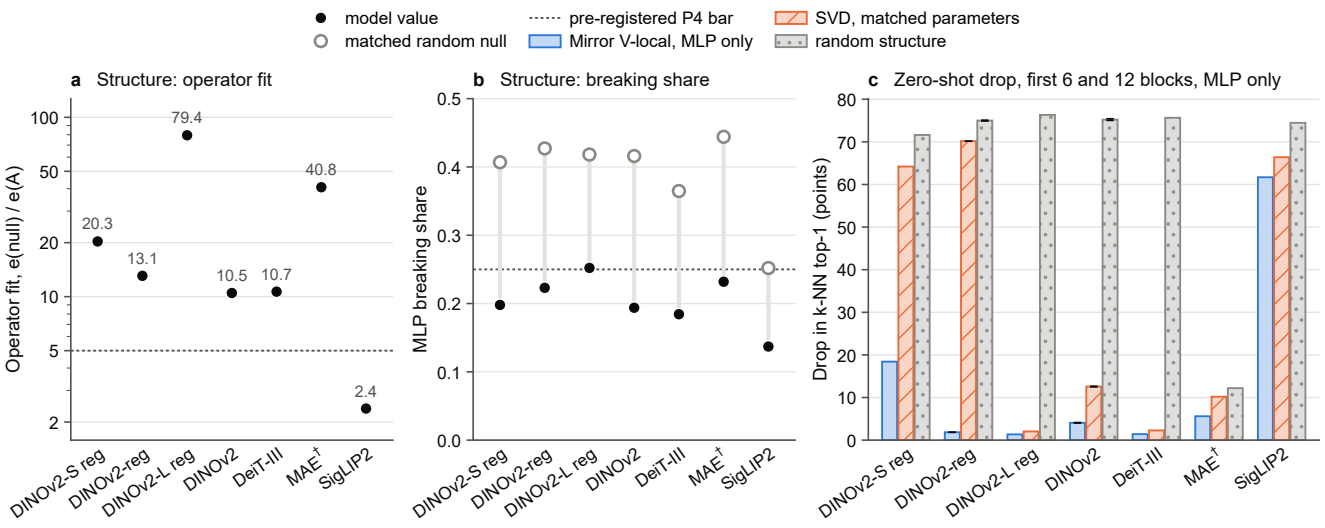

model value
matched random null
pre-registered P4 bar
Mirror V-local, MLP only
SVD, matched parameters
random structure
a Structure: operator fit
Operator fit, e(null) / e(A)
100
50
20
10
5
2
79.4
40.8
20.3
13.1
10.5
10.7
2.4
DINOv2-S reg
DINOv2-reg
DINOv2-L reg
DINOv2
DeiT-III
MAE†
SigLIP2
b Structure: breaking share
MLP breaking share
0.5
0.4
0.3
0.2
0.1
0.0
DINOv2-S reg
DINOv2-reg
DINOv2-L reg
DINOv2
DeiT-III
MAE†
SigLIP2
c Zero-shot drop, first 6 and 12 blocks, MLP only
Drop in k-NN top-1 (points)
80
70
60
50
40
30
20
10
0
DINOv2-S reg
DINOv2-reg
DINOv2-L reg
DINOv2
DeiT-III
MAE†
SigLIP2